Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The maximum value of $f(x) = 2\sin x + \sin 2x$, in the interval $\left[0, \frac{3\pi}{2}\right]$ is:

(A) $\sqrt{2}+1$

(B) $2\sqrt{3}$

(C) $\frac{3\sqrt{3}}{2}$

(D) $\sqrt{3}$

Q.2

The minimum value of $27\tan^2\theta + 3\cot^2\theta$ is

(A) 9

(B) 18

(C) 27

(D) 30

Q.3

If $u = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta} + \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$ where $a, b > 0$. Then the difference between the maximum and minimum values of u^2 is given by

(A) $(a+b)^2$

(B) $2\sqrt{a^2+b^2}$

(C) $2\left(a^2+b^2\right)$

(D) $(a-b)^2$

Q.4

The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is

(A) 4

(B) 3

(C) 2

(D) 1

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Q.5

$\tan x + \frac{1}{2}\tan\frac{x}{2} + \frac{1}{2^2}\tan\frac{x}{2^2} + \ldots + \frac{1}{2^{n-1}}\tan\frac{x}{2^{n-1}}$ is equal to

(A) $\frac{1}{2^n}\cot\frac{x}{2^n} - 2\cot 2x$

(B) $\frac{1}{2^{n-1}}\cot\left(\frac{x}{2^{n-1}}\right) - 2\cot 2x$

(C) $\tan\frac{2^n-1}{2^{n-1}}x$

(D) $2\cot 2x - \frac{1}{2^{n-1}}\cot\frac{x}{2^{n-1}}$

Q.6

If $\sin\theta + \cos\theta = a$ and $\cos\theta - \sin\theta = b$, then $\sin\theta(\sin\theta - \cos\theta) + \sin^2\theta\left(\sin^2\theta - \cos^2\theta\right) + \sin^3\theta\left(\sin^3\theta - \cos^3\theta\right) + \ldots$ is equal to

(A) $\frac{1-ab}{1+ab}$

(B) $\frac{1-a^2}{3-a^2}$

(C) $\frac{1-ab}{1+ab} + \frac{1-a^2}{3-a^2}$

(D) $\frac{1+ab}{1-ab} - \frac{a^2-1}{3-a^2}$

Q.7

If $0 < \theta < \pi/2$, number of solutions of $\sum_{m=1}^{6}\operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is

(A) 0

(B) 1

(C) 2

(D) 4

Q.8

The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are:

(A) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{12}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(B) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

(C) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Q.9

If $4n\alpha = \pi$, then the value of $\tan\alpha\tan 2\alpha\tan 3\alpha\ldots\tan(2n-1)\alpha$ is

(A) -1

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(B) 0

(C) 1

(D) none of these

Q.10

The number of solutions of $\sin\theta + 2\sin 2\theta + 3\sin 3\theta + 4\sin 4\theta = 10, 0 < \theta < \pi$ is

(A) 0

(B) 1

(C) 2

(D) 4

Q.11

$\sin x, \sin 2x, \sin 3x$ are in A.P. if

(A) $x = 165^\circ$

(B) $x = 195^\circ$

(C) $x = 15^\circ$

(D) $x = 2n\pi \quad n \in I.$

Q.12

The maximum value of $\sin\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$ is in the interval $\left(0, \frac{\pi}{2}\right)$ if the value of x is

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{12}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{4}$

Q.13

If an angle B is complement of an angle A, what are the greatest and least values of $\cos A \cos B$ respectively?

(A) $0, -\frac{1}{2}$

(B) $\frac{1}{2}, -1$

(C) 1,0

(D) $\frac{1}{2}, -\frac{1}{2}$

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Q.14

If $\alpha + \beta + \gamma = \pi$ then the minimum value of $\cos A + \cos B + \cos C$

(A) is zero

(B) is positive

(C) lies between -2 and -3

(D) is -3

Q.15

The value of $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \sin^2 20^\circ + \ldots + \sin^2 90^\circ$ is

(A) 7

(B) 8

(C) 9

(D) $\frac{19}{2}$

Answers & Solutions

Q.1 Answer:

$\frac{3\sqrt{3}}{2}$

Solution:

$$f(x) = 2\sin x + \sin 2x = 2\sin x + 2\sin x \cos x$$
$$= 2\sin x(1 + \cos x)$$
$$f'(x) = 2\cos x(1 + \cos x) - 2\sin x \sin x$$
$$= 2\left[\cos x + \cos^2 x - \left(1 - \cos^2 x\right)\right]$$
$$= 2\left[2\cos^2 x + \cos x - 1\right]$$
$$= 2(\cos x + 1)(2\cos x - 1)$$

Now, $f'(x) = 0 \Rightarrow x = \pi/3, \pi \quad (\because 0 \le x \le 3\pi/2)$

We have $f(0) = 0$, **of** $\left(\frac{3\pi}{2}\right) = -2$

$f\left(\frac{\pi}{3}\right) = 2\left(\frac{\sqrt{3}}{2}\right) + \frac{\sqrt{3}}{2} = \frac{3}{2}\sqrt{3}$

and $f(\pi) = 0, f\left(\frac{3\pi}{2}\right) = -2$

Thus, maximum value is $\frac{3}{2}\sqrt{3}$.

Q.2 Answer:

18

Solution:

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A.M. $\geq$ G.M.

$\Rightarrow \quad \frac{27\tan^2\theta+3\cot^2\theta}{2} \geq \sqrt{27\tan^2\theta \cdot 3\cot^2\theta}$

$\Rightarrow \quad 27\tan^2\theta + 3\cot^2\theta \geq 18$

Q.3 Answer:

$(a-b)^2$

Solution:

We have

$$u = \sqrt{\frac{a^2}{2}(1+\cos 2\theta) + \frac{b^2}{2}(1-\cos 2\theta)} + \sqrt{\frac{a^2}{2}(1-\cos 2\theta) + \frac{b^2}{2}(1+\cos 2\theta)}$$

$$= \sqrt{\frac{a^2+b^2}{2} + \frac{a^2-b^2}{2}\cos 2\theta} + \sqrt{\frac{a^2+b^2}{2} - \frac{a^2-b^2}{2}\cos 2\theta}$$

Squaring we get

$$u^2 = a^2+b^2+2\sqrt{\left(\frac{a^2+b^2}{2}\right)^2 - \left(\frac{a^2-b^2}{2}\right)^2\cos^2 2\theta}$$

$$\text{Thus max } \left(u^2\right) = a^2+b^2+a^2+b^2 = 2\left(a^2+b^2\right)$$

$$\text{and min } \left(u^2\right) = a^2+b^2+2\sqrt{\left(\frac{a^2+b^2}{2}\right)^2 - \left(\frac{a^2-b^2}{2}\right)^2}$$

$$= a^2+b^2+2ab = (a+b)^2$$

So the required difference

$= 2\left(a^2+b^2\right) - (a+b)^2 = (a-b)^2.$

Q.4 Answer:

4

Solution:

$$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$$

$$= \tan 9^\circ + \tan(90^\circ - 9) - (\tan 27^\circ + \tan(90^\circ - 27^\circ))$$

$$= \tan 9^\circ + \cot 9^\circ - (\tan 27^\circ + \cot 27^\circ)$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = \frac{2\times 4}{\sqrt{5}-1} - \frac{2\times 4}{\sqrt{5}+1} = 4$$

Q.5 Answer:

$\frac{1}{2^{n-1}}\cot\left(\frac{x}{2^{n-1}}\right) - 2\cot 2x$

Solution:

We can write $\tan x = \frac{\tan^2 x - 1 + 1}{\tan x}$

$= \cot x - 2\cot 2x$

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so $\tan x + \frac{1}{2}\tan\frac{x}{2} + \frac{1}{2^2}\tan\frac{x}{2^2} + \cdots$
$+\frac{1}{2^{n-1}}\tan\frac{x}{2^{n-1}}$
$= \cot x - 2\cot 2x + \frac{1}{2}\left(\cot\frac{x}{2} - 2\cot x\right)$
$\quad + \frac{1}{2^2}\left(\cot\frac{x}{2^2} - 2\cot\frac{x}{2}\right)$
$\quad + \cdots + \frac{1}{2^{n-1}}\left(\cot\frac{x}{2^{n-1}} - 2\cot\frac{x}{2^{n-2}}\right)$
$= \frac{1}{2^{n-1}}\cot\left(\frac{x}{2^{n-1}}\right) - 2\cot 2x$

Q.6 Answer:

$\frac{1-ab}{1+ab} + \frac{1-a^2}{3-a^2}$

Solution:

We have $\sin^2\theta + \sin^4\theta + \sin^6\theta + \cdots$
$-\left(\sin\theta\cos\theta + \sin^2\theta\cos^2\theta + \sin^3\theta + \cos^3\theta + \cdots\right)$
$= \frac{\sin^2\theta}{1-\sin^2\theta} - \frac{\sin\theta\cos\theta}{1-\sin\theta\cos\theta}$
$= \frac{1-\cos 2\theta}{1+\cos 2\theta} - \frac{\sin 2\theta}{(2-\sin 2\theta)}$
$= \frac{1-ab}{1+ab} + \frac{1-a^2}{3-a^2}$

Q.7 Answer:

2

Solution:

We can write
$\frac{1}{\sqrt{2}}\sum_{m=1}^{6}\frac{1}{\sin\left(\theta+\frac{(m-1)\pi}{4}\right)\sin\left(\theta+\frac{m\pi}{4}\right)} = 4$
$\Rightarrow \quad \sum_{m=1}^{6}\frac{\sin\left[\left(\theta+\frac{m\pi}{4}\right)-\left(\theta+\frac{(m-1)\pi}{4}\right)\right]}{\sin\left(\theta+\frac{(m-1)\pi}{4}\right)\sin\left(\theta+\frac{m\pi}{4}\right)} = 4$
$\Rightarrow \quad \sum_{m=1}^{6}\left[\cot\left(\theta+\frac{(m-1)\pi}{4}\right) - \cot\left(\theta+\frac{m\pi}{4}\right)\right] = 4$
$\Rightarrow \quad \cot\theta - \cot\left(\theta+\frac{6\pi}{4}\right) = 4$
$\Rightarrow \cot\theta + \tan\theta = 4$
$\Rightarrow \sin 2\theta = 1/2$
$\Rightarrow \quad 2\theta = \pi/6, 5\pi/6 \Rightarrow \theta = \pi/12, 5\pi/12$

Q.8 Answer:

$\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Solution:

We have
$\sin\theta + \sin(4\theta) + \sin(7\theta) = 0$
$\Rightarrow 2\sin(4\theta)\cos(3\theta) + \sin(4\theta) = 0$
$\Rightarrow \sin(4\theta)[2\cos(3\theta) + 1] = 0$
$\Rightarrow \quad \sin(4\theta) = 0$ or $\cos(3\theta) = -1/2 = \cos(2\pi/3)$

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$\Rightarrow \quad \theta = \pi/4, \pi/2, 3\pi/4$ or $3\theta = 2\pi/3, 2\pi \pm 2\pi/3$

$\Rightarrow \quad \theta = \frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{4\pi}{9}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Q.9 Answer:

1

Solution:

$\tan(2n-1)\alpha = \tan(\pi/2 - \alpha) = \cot\alpha$

$\tan(2n-2)\alpha = \cot 2\alpha$ **and so on. So the required value is 1.**

Q.10 Answer:

0

Solution:

Given equation is possible if $\sin\theta = \sin 2\theta = \sin 3\theta$

$= \sin 4\theta = 1$ **for some** $\theta, 0 < \theta < \pi$ **which is not possible.**

Q.11 Answer:

$x = 2n\pi \quad n \in I.$

Solution:

$\sin x, \sin 2x, \sin 3x$ **are in A.P.**

If $2\sin 2x = \sin x + \sin 3x = 2\sin 2x \cos x$

or if $2\sin 2x(1 - \cos x) = 0$

or if either $\sin 2x = 0$ **or** $\cos x = 1$

which is satisfied by $x = 2n\pi$ **for which** $\cos x = 1$

Q.12 Answer:

$\frac{\pi}{12}$

Solution:

Let, $y = \sin\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$

$= \sqrt{2}\left[\frac{1}{\sqrt{2}}\sin\left(x + \frac{\pi}{6}\right) + \frac{1}{\sqrt{2}}\cos\left(x + \frac{\pi}{6}\right)\right]$

$= \sqrt{2}\left[\sin\frac{\pi}{4}\sin\left(x + \frac{\pi}{6}\right) + \cos\frac{\pi}{4}\cos\left(x + \frac{\pi}{6}\right)\right]$

$= \sqrt{2}\left[\cos\left(x + \frac{\pi}{6} - \frac{\pi}{4}\right)\right] = \sqrt{2}\left[\cos\left(x - \frac{\pi}{12}\right)\right]$

$\Rightarrow \quad x - \frac{\pi}{12} = 0 \quad [\because y$ **to be max.** $]$

$\Rightarrow \quad x = \frac{\pi}{12}$

Q.13 Answer:

$\frac{1}{2}, -\frac{1}{2}$

Solution:

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Since, A and B are complementary angles, then $\mathrm{A}+\mathrm{B}=90^{\circ}$
Now, $\cos A \cos B=\cos A \cos \left(90^{\circ}-A\right)$
$=\cos \mathrm{A} \sin \mathrm{A}=\frac{1}{2} \sin 2 \mathrm{~A}$
Since, $-1 \leq \sin 2 \mathrm{~A} \leq 1$
Hence, $-\frac{1}{2} \leq \frac{1}{2} \sin 2 \mathrm{~A} \leq \frac{1}{2}$.
Thus, greatest and least values of $\cos \mathrm{A} \cos \mathrm{B}$
are $\frac{1}{2}$ and $-\frac{1}{2}$.

Q.14 Answer:

is -3

Solution:

For all $x, \cos x \geq -1$
$\therefore \cos A+\cos B+\cos C \geq -3$ and equality
holds if $\cos \mathrm{A}=\cos \mathrm{B}=\cos \mathrm{C}=-1$, which can be attained
if $\mathrm{A}=\pi, \mathrm{B}=\pi, \mathrm{C}=-\pi$

Q.15 Answer:

$\frac{19}{2}$

Solution:

$\sin^2 5^{\circ}+\sin^2 10^{\circ}+\sin^2 15^{\circ}+\ldots \ldots \ldots+\ldots \ldots \ldots$
$\sin^2 75^{\circ}+\sin^2 80^{\circ}+\sin^2 85^{\circ}+\sin^2 90^{\circ}$
$\Rightarrow \sin^2 5^{\circ}+\sin^2 10^{\circ}+\sin^2 15^{\circ}+\ldots \ldots \sin^2$
$(90-15^{\circ})+\sin^2(90-10)+\sin^2(90-5)+1$
$\Rightarrow \quad \sin^2 5+\sin^2 10+\sin^2 15^{\circ}+\ldots \ldots \ldots+\cos^2 15^{\circ}$
$+\cos^2 10^{\circ}+\cos^2 5^{\circ}+1$
$\Rightarrow (1+1+1+\ldots .8$ **times** $)+\sin^2 45^{\circ}+1$
$\Rightarrow \quad 8+\frac{1}{2}+1=\frac{19}{2}$

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